

9.A Periodicity and Ionic Bonding I

Core and Valence Electrons; Effective Nuclear Charge. In this chapter, we will examine several patterns and trends in the properties of the elements. Trends observed left to right across a period on the periodic table tend to repeat themselves; as a result, they are called “periodic trends.” Periodic trends can be readily understood and rationalized using atomic electron configurations and the properties of electrons in the outermost shell of the atom. Here, we’ll introduce some terminology to distinguish between inner and outer shells and see that the outermost electrons in an atom do not “feel” all the nuclear charge—and the amount of nuclear charge they do feel has a profound effect on the properties of the element.

- **Core electrons** are in the inner shells of the atom; core electrons remain close to the nucleus and do not participate in bonding or chemical reactions.
 - Core electrons occupy completely full shells.
 - They just follow the nucleus around; you can think of them as “embedded in” the element symbol in molecular structures.
- **Valence electrons** are in the outermost shell of the atom.
 - Valence electrons are engaged in bonding and electron transfer.
- For the main-group elements, the number of valence electrons in the neutral atom can be inferred from the group number.

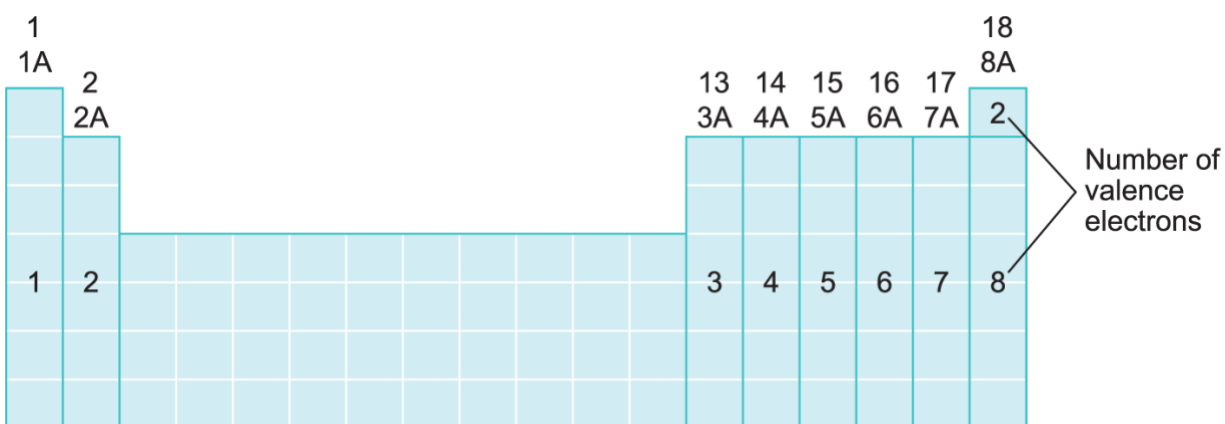
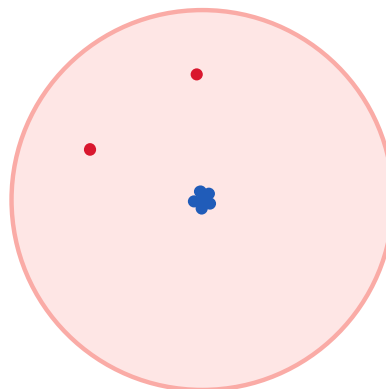


Figure 9.1. Valence electron counts for the main-group elements.

- Several important properties of the elements themselves depend on the energies of valence electrons. These energies are determined by two electrostatic factors...

- Electron-nucleus attraction.** Negatively charged electrons are attracted to positively charged protons in the nucleus.
- Electron-electron repulsion.** Negatively charged electrons repel one another.



- Interactions between the electrons and nucleus follow electrostatic principles...

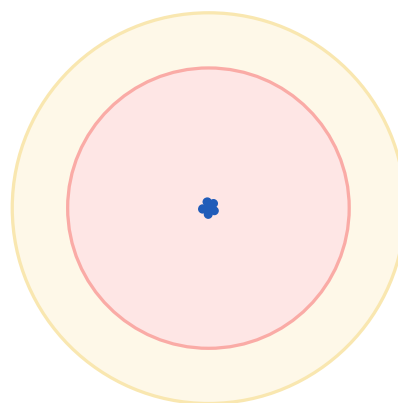
$$E \propto \frac{q_1 q_2}{r}$$

- The valence electrons do not “feel” the full attractive force of the nucleus due to repulsion from other valence electrons and (more important) core electrons.

- Effective nuclear charge (Z_{eff})** is the net positive charge felt by an electron after accounting for electron-electron repulsion.

- In this context, electron-electron repulsion that reduces the value of Z_{eff} is called **shielding**.

- Z_{eff} for the outermost (“last”) valence electron is most interesting, since this is the most energetic electron.



- Slater’s rules** are a set of conventions for determining the value of the shielding constant S for the last valence electron. Only the numbers of core (N_{core}) and valence electrons (N_{val}) are used; distinctions are not made between different core shells.

$$S = 0.85N_{core} + 0.35(N_{val} - 1)$$

$$Z_{eff} = Z - S$$

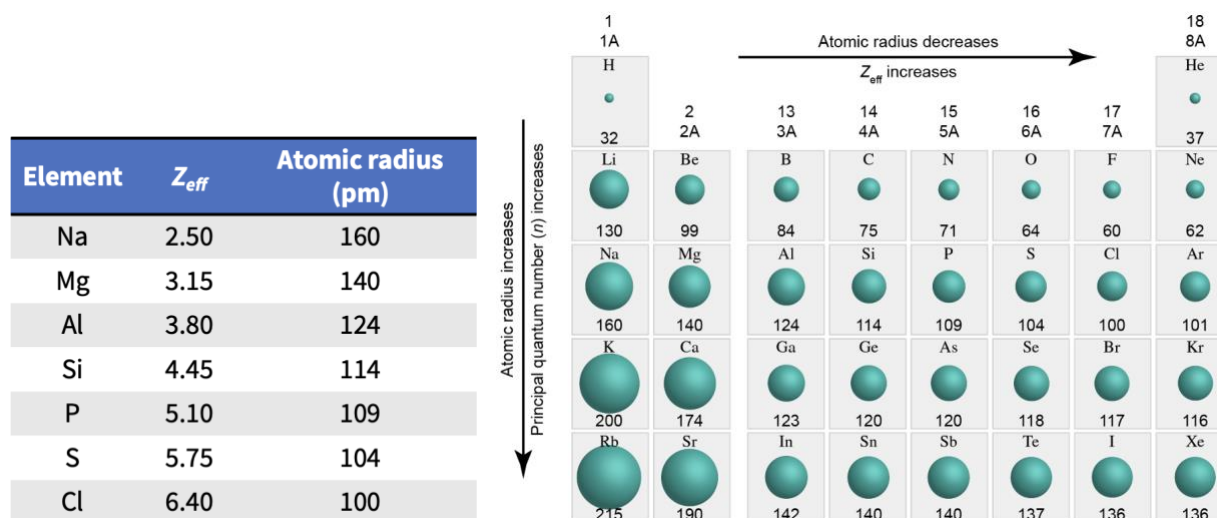


Example. Calculate the effective nuclear charge felt by the outermost valence electron in a neutral neon atom.

Choose a main group element on the [CHEM 1211K Periodic Trends](#) spreadsheet. Add the number of valence electrons and the number of core electrons for your element; Z_{eff} is calculated automatically from these numbers using Slater's rules. Do you notice any trends in Z_{eff} across the period of your element?

Atomic and Ionic Radii. Reflecting on the quantum nature of electrons in atoms, we might conclude that the atom has no well-defined “size”—wave functions trail off to infinity. However, assuming we establish some generally acceptable cutoff for orbital density (there are a few different valid approaches), we can measure a radius for each atom and identify patterns and trends. Atomic and ionic radii are related in a very deep way to Z_{eff} . For the remainder of this chapter, we'll see that Z_{eff} is the root cause of all periodic trends!

- **Atomic radius** increases down a group and generally decreases from left to right across a period.
- Note that atomic radius is *negatively* correlated with Z_{eff} across a period.



9.1. Atoms A and B are in the same period and atom A has a smaller atomic radius than atom B. Which statement *best* explains why this is true?

- A. The valence electrons of atom B are more strongly attracted to its nucleus than are the valence electrons of atom A to its nucleus.
- B. Atom A has fewer protons than atom B.
- C. Z_{eff} for A (for the outermost valence electron) is greater than Z_{eff} for B.
- D. Atom A has more core electrons than B.
- E. Atom B has more valence electrons than A.

9.2. Arrange these atoms in order of increasing radius:

- A. Nitrogen
- B. Oxygen
- C. Sodium
- D. Phosphorus

Enter your response as a string of letters from smallest to largest, such as "ABCD."

- All cations have smaller radii than their corresponding neutral atoms.
 - The neutral atom loses electrons → shielding decreases → Z_{eff} increases
- All anions have larger radii than their corresponding neutral atoms.
 - The neutral atom gains electrons → shielding increases → Z_{eff} decreases

1 1A		2 2A		16 6A		17 7A	
Li	Li ⁺	Be	Be ²⁺	O	O ²⁻	F	F ⁻
							
130	76	99	31	64	140	72	133
Na	Na ⁺	Mg	Mg ²⁺	S	S ²⁻	Cl	Cl ⁻
							
160	102	140	72	104	184	100	181
K	K ⁺	Ca	Ca ²⁺	Se	Se ²⁻	Br	Br ⁻
							
200	138	174	100	118	198	117	196
Rb	Rb ⁺	Sr	Sr ²⁺	Te	Te ²⁻	I	I ⁻
							
215	152	190	118	137	221	136	220

- When electrons are lost or gained from a neutral atom to form an ion, the numbers of protons and core electrons remain constant, but the number of valence electrons changes. Z_{eff} changes as a result.
 - For example, consider neutral Ca versus Ca^{2+} .
- Cations and anions decrease in radius left to right across a period and increase in radius down a group, just like neutral atoms.

Chemical symbol K^+
Ionic radius (pm) 138
Relative size

1 1A	2 2A	13 3A	15 5A	16 6A	17 7A
Li^+ 76	Be^{2+} 31	B^{3+} 23	N^{3-} 146	O^{2-} 140	F^- 133
Na^+ 102	Mg^{2+} 72	Al^{3+} 54	P^{3-} 212	S^{2-} 184	Cl^- 181
K^+ 138	Ca^{2+} 100	Ga^{3+} 62	As^{3-} 222	Se^{2-} 198	Br^- 196
Rb^+ 152	Sr^{2+} 118	In^{3+} 81		Te^{2-} 221	I^- 220

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9.3. Arrange these ions in order of increasing radius:

- Ca^{2+}
- Cl^-
- S^{2-}
- K^+

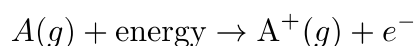
Enter your response as a string of letters from smallest to largest, such as "ABCD."

9.B Periodicity and Ionic Bonding II

Revisit your element on the [CHEM 1211K Periodic Trends](#) spreadsheet. Use [ptable.com](#) to add the empirical atomic radius (in pm), first ionization energy (in eV), and electron affinity (in eV) for the element. The data will be graphed automatically against atomic number. What patterns and trends do you notice?

Ionization Energy. In oxidation-reduction or redox reactions, electrons are transferred from one atom or molecule to another. The energetics of redox reactions are related to the ease with which electrons are removed from or added to atoms. Since the outermost (“last”) valence electron is the highest in energy and Z_{eff} is a measure of the “tightness” with which this electron is bound in its atom, the energetics of electron transfer come down to Z_{eff} . We begin by focusing on the energy required to remove an electron from a gaseous atom, the so-called “ionization energy.”

- Energies involved in adding an electron to or removing an electron from the atom follow periodic trends.
- **Ionization energy** is the energy input required to remove an electron from a gaseous atom to produce a gaseous cation and free electron.



- Energy is *always* required to ionize a neutral atom: ionization energy is positive.
- Typical units are kilojoules per mole (kJ/mol) or electron-Volts (eV) per atom.
- Ionization energy is also measurable for molecules.
- Consider the energy levels of the hydrogen atom from the last chapter. We have seen the ionization energy of H(g) before!

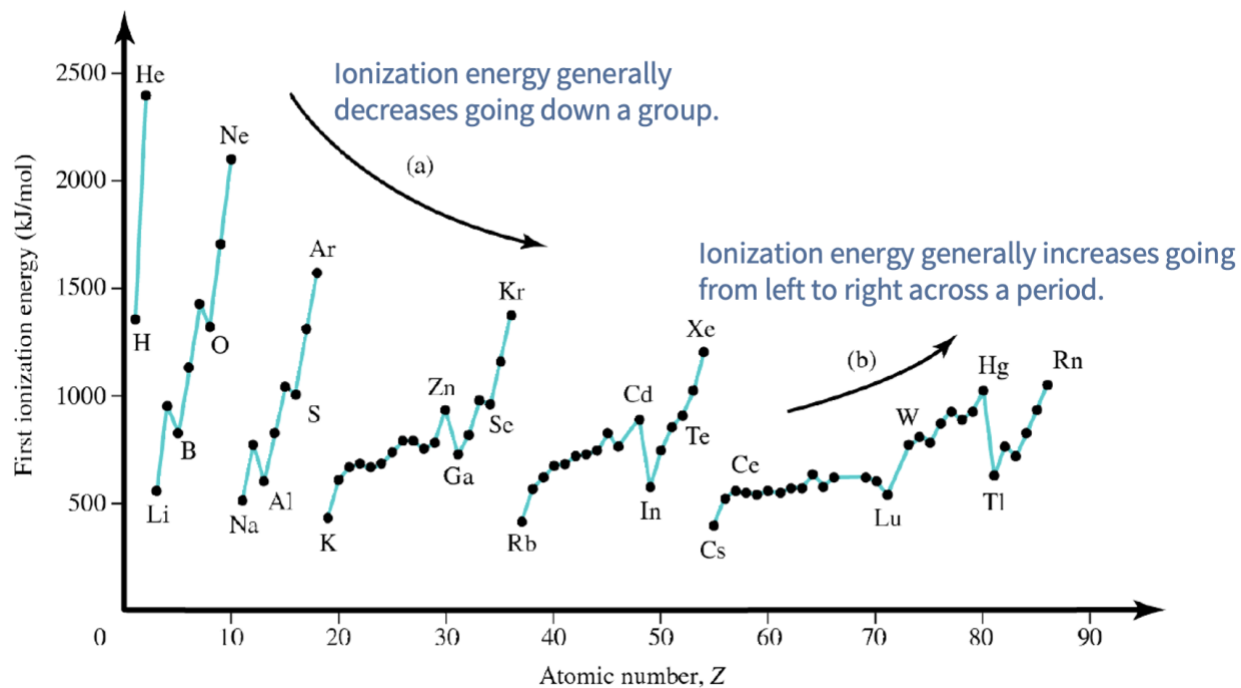


Figure 9.7. Ionization energies of the elements.

- Ionization energy and atomic radius are negatively correlated...why?

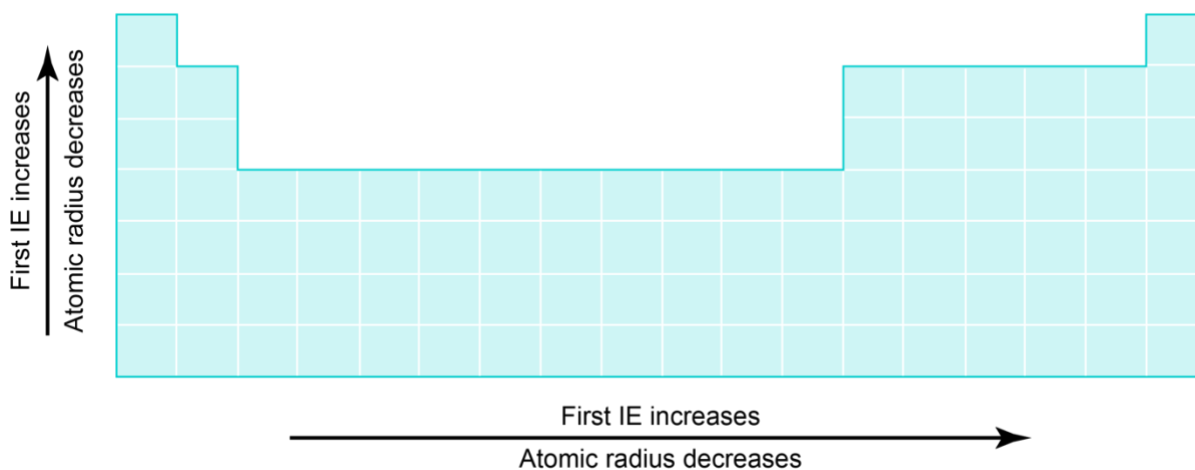


Figure 9.8. Trends in first ionization energy and atomic radius.

- Some exceptions to the general trend in ionization energy occur when the outermost valence electron is in a subshell by itself. It is easier to remove this electron than trends predict.

Example. Explain why the ionization energy of boron (B, 8.30 eV) is lower than the ionization energy of beryllium (Be, 9.32 eV).

- Other exceptions involve elements containing only one set of paired electrons in the valence p subshell (group 16). Again, it is easier to remove this electron than trends predict.

Example. Explain why the ionization energy of oxygen (13.6 eV) is lower than the ionization energy of nitrogen (14.5 eV).

9.4. Which statement is *true* regarding first ionization energy? (“First” refers to loss of the first electron from the neutral atom.)

- A. It represents the energy released when an electron is removed from a gas phase atom.
- B. The first ionization energy for lithium is greater than that for fluorine.
- C. First ionization energy corresponds to the process: $X(g) + e^- \rightarrow X^-(g) + IE$.
- D. The first ionization energy for lithium should be less than that for sodium because sodium has fewer core electrons.
- E. The first ionization energies for noble gases are very large because of their very stable electron configurations.

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- Higher ionization energies involve the removal of a second electron, third, fourth, etc.
 - Because Z_{eff} increases significantly with each electron lost, $IE_1 < IE_2 < IE_3 < \dots$
 - IE for the first core electron lost is ***much*** larger than the previous IE! See ptable.com.

9.5. An unknown main group element in the fourth period experiences a large jump between the sixth and seventh ionization energies. What is the identity of the element? Enter its chemical symbol.

Electron Affinity. Electrons would almost always rather be inside atoms than outside. (Wouldn't you rather be inside Mercedes-Benz Stadium than outside?) As a result, energy is typically released when a free electron is "captured" by a neutral atom to form an anion. The amount of energy released is referred to as the "electron affinity" of the element. We'll once again see that Z_{eff} is central to explaining observed trends in electron affinity, but we'll also note some important exceptions that can be rationalized by carefully considering electron configuration in the neutral atom and anion.

- **Electron affinity** is the change in energy when a neutral atom in the gas phase captures a free electron and forms a gaseous anion.



- Electron capture is *NOT* the reverse of ionization to a cation as shown above! It is conceptually distinct.
- Electron capture is usually exothermic due to positive Z_{eff} in the valence shell...
- ...however, when the valence shell is full (group 18) or even a valence subshell is full (groups 2 and 12), electron capture is endothermic.

- Electron affinity predictably becomes less exothermic moving down a group, since the new electron is accepted into a shell farther from the nucleus.
- Electron affinity *generally* becomes more negative left to right across a period (ignoring the noble gases).

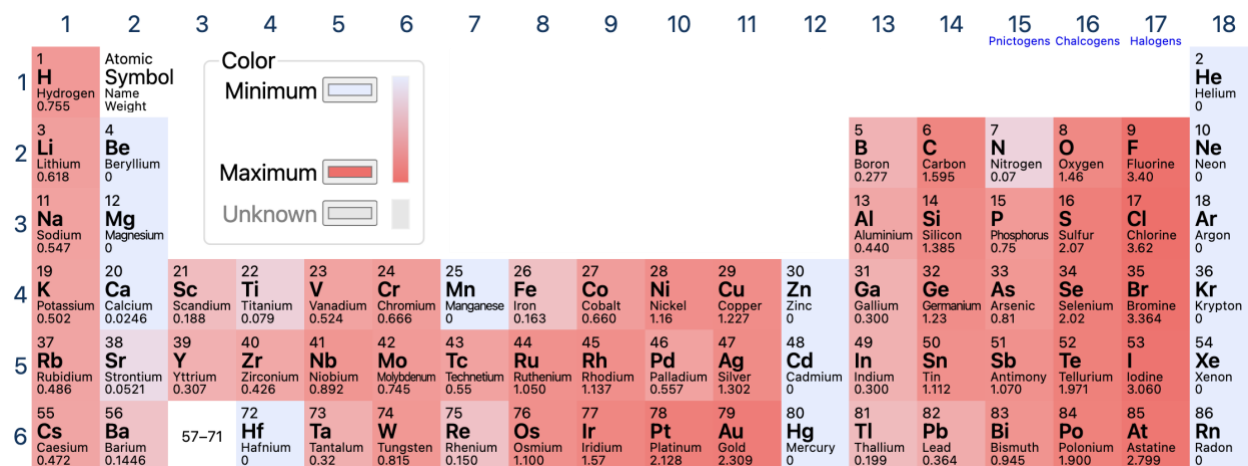
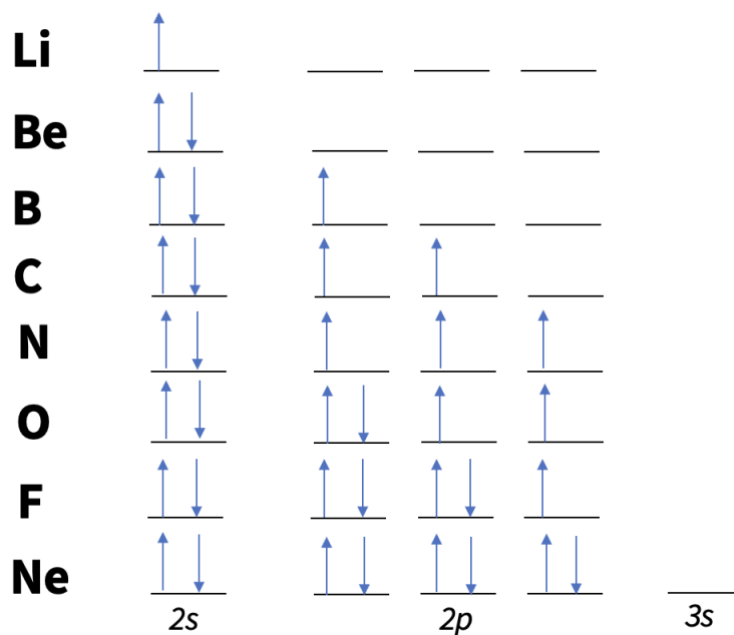


Figure 9.10. Electron affinities of the elements in electron-Volts (eV). A positive value indicates the quantity of energy released per electron captured; zero indicates endothermic electron capture.

- Why the numerous exceptions? Consider the electron configuration after accepting a new electron. For which elements are there “energetic penalties” associated with accepting another electron?

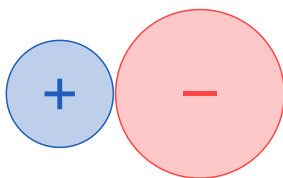


9.6. Rank the following elements from lowest (1) to highest (5) electron affinity. We will treat electron affinities as positive values—if it helps, read it as “affinity for electrons.” Greater affinity is associated with a higher number, as it should be.

- A. Li
- B. F
- C. N
- D. O
- E. Na

Ionic Bonding and Lattice Energy. Here, we begin to transition from thinking about elements and atomic properties to thinking about compounds and molecular properties. We begin with an overview of the nature of ionic bonding. Lattice energy is a measure of ionic bond strength that follows trends involving charge and ionic radii.

- An **ionic bond** is the electrostatic attraction between oppositely charged ions.



- Ionic compounds contain an essentially infinite three-dimensional **lattice** of cations and anions.

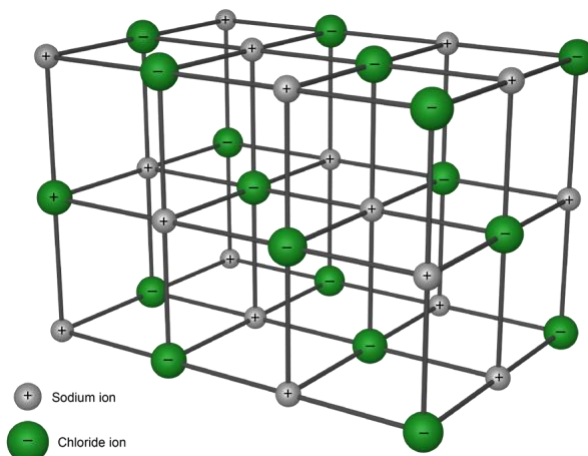
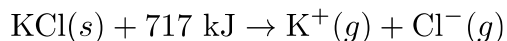
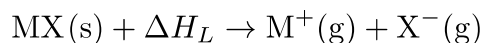
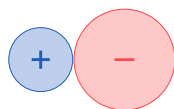


Figure 9.12. Ionic lattice in sodium chloride.

- Ionic bond strength can be measured as **lattice energy (ΔH_L)**, the energy input required to “blast” 1 mole of an ionic lattice into gaseous ions.¹



- Trends in lattice energy can be accounted for by **Coulomb's law**.



$$\Delta H_L \propto \left| \frac{q_1 q_2}{r} \right|$$

- Smaller ions are associated with more stable lattices and higher lattice energy (shorter r).
- Charges greater in magnitude are associated with more stable lattices and higher lattice energy (greater $q_1 q_2$).

9.7. Order the following compounds from greatest (1) to smallest (5) lattice energy. Enter your response as a string of letters from largest to smallest lattice energy, such as “ABCDE.”

- A. CsBr
- B. SrO
- C. CrN
- D. MgO
- E. Na₂S

¹ Your textbook may report and discuss lattice energy as a negative value, the enthalpy change when gaseous ions come together to form 1 mole of the ionic solid. To avoid confusion involving “greater than” and “less than,” we will follow the prevailing convention and refer to lattice energies using positive values.

- Lattice energies are not directly measured, but instead are calculated by applying Hess's law to a series of connected processes called a **Born-Haber cycle**.

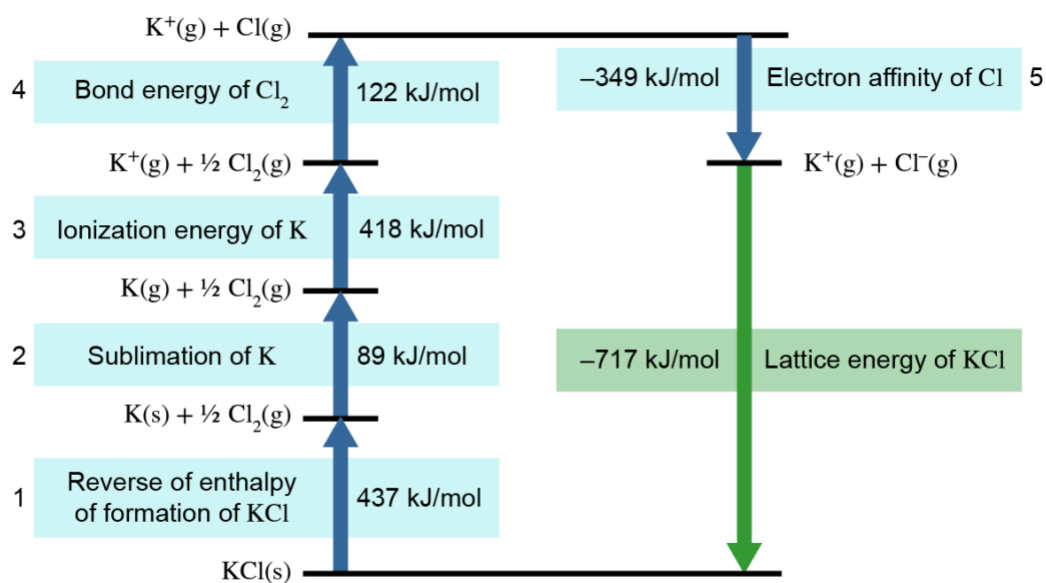


Figure 9.14. Born-Haber cycle for potassium chloride.

Chapter Summary. Some of the amazing things we can now do with periodicity and ionic bonding...

- Determine the number of valence electrons in a neutral atom of a main group element.
- Calculate the effective nuclear charge felt by an electron in the valence subshell (Z_{eff}) given the identity of an atom or ion, using Slater's rules or related systems.
- Apply periodic trends in atomic and ionic radii to predict the relative sizes of atoms and/or ions; rationalize these trends using electron configuration and Z_{eff} .
- Apply periodic trends in ionization energy to predict the relative ionization energies of a series of elements.
- Apply periodic trends in electron affinity to predict the relative electron affinities of a series of elements.
- Rationalize trends in ionization energy and electron affinity (and exceptions) using electron configuration and Z_{eff} .
- Apply periodic trends in ionic radii and charges of monatomic ions to predict the relative lattice energies of a series of binary ionic compounds.